

Ex 5 Data Collection and Initial Exploration

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from google.colab import files
uploaded = files.upload()
df = pd.read_csv("education.csv")
print(df)

print("_____")
df.info()
print("_____")
print(df.describe())
print("_____")
sns.pairplot(df)
plt.show()

df.hist(bins=10,figsize=(12,10))
plt.show()
```

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Saving education.csv to education (5).csv

	Student_ID	Name	Age	Gender	Course	Department	Year	Marks	\
0	1	Asha	19	F	BCA	Computer Science	1	85	
1	2	Ravi	20	M	BSc	Mathematics	2	78	
2	3	Kumar	18	M	BCom	Commerce	1	92	
3	4	Priya	21	F	BCA	Computer Science	3	88	
4	5	Arun	19	M	BSc	Physics	2	74	
5	6	Divya	20	F	BCom	Commerce	1	81	
6	7	Karthik	18	M	BCA	Computer Science	3	95	
7	8	Meena	19	F	BSc	Mathematics	2	69	
8	9	Vijay	21	M	BCom	Commerce	1	83	
9	10	Anitha	20	F	BCA	Computer Science	2	90	
10	11	Rahul	19	M	BSc	Physics	3	77	
11	12	Sneha	18	F	BCom	Commerce	1	86	
12	13	Surya	20	M	BCA	Computer Science	2	91	
13	14	Pooja	21	F	BSc	Mathematics	3	73	
14	15	Manoj	19	M	BCom	Commerce	2	89	
15	16	Kavya	20	F	BCA	Computer Science	1	84	
16	17	Ajay	18	M	BSc	Physics	3	76	
17	18	Nisha	19	F	BCom	Commerce	2	93	
18	19	Rohit	21	M	BCA	Computer Science	1	87	
19	20	Deepa	20	F	BSc	Mathematics	3	80	

```

...      Attendance      City
0         92      Chennai
1         85      Madurai
2         96  Coimbatore
3         89      Chennai
4         78       Salem
5         91       Trichy
6         94      Chennai
7         75      Madurai
8         88  Coimbatore
9         93      Chennai
10        82       Salem
11        90       Trichy
12        95      Chennai
13        79      Madurai
14        87  Coimbatore
15        86      Chennai
16        81       Salem
17        94       Trichy
18        90      Chennai
19        84      Madurai

```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 20 entries, 0 to 19
```

```
Data columns (total 10 columns):
```

#	Column	Non-Null Count	Dtype
0	Student_ID	20 non-null	int64
1	Name	20 non-null	object
2	Age	20 non-null	int64

```

...
3  Gender      20 non-null    object
4  Course      20 non-null    object
5  Department  20 non-null    object
6  Year        20 non-null    int64
7  Marks       20 non-null    int64
8  Attendance  20 non-null    int64
9  City        20 non-null    object
dtypes: int64(5), object(5)
memory usage: 1.7+ KB

```

	Student_ID	Age	Year	Marks	Attendance
count	20.00000	20.00000	20.00000	20.00000	20.00000
mean	10.50000	19.50000	1.95000	83.55000	87.45000
std	5.91608	1.051315	0.825578	7.337539	6.082546
min	1.00000	18.00000	1.00000	69.00000	75.00000
25%	5.75000	19.00000	1.00000	77.75000	83.50000
50%	10.50000	19.50000	2.00000	84.50000	88.50000
75%	15.25000	20.00000	3.00000	89.25000	92.25000
max	20.00000	21.00000	3.00000	95.00000	96.00000



